Salih Mithat Kaya

Alireza Siadatan

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Bonus Project 3 - 2x4 Decoder Using NAND Gates

The purpose of this report is to design and implement a 2x4 decoder only using basic NAND logic gates to act as a real decoder chip.

Part 1: Setup

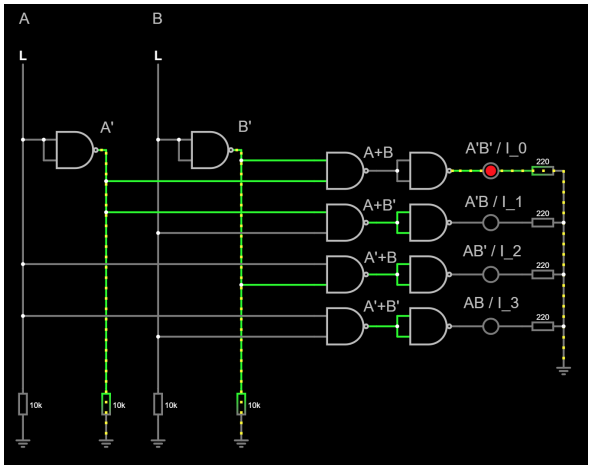
This projects only requires three Quad 2 input NAND gate IC chips, as we will need two logic gates to invert A and B respectively, and need two logic gates per I_out computation.

Besides the IC chips, a breadboard, 5V power supply module, two 10k pulldown resistors, four LED's, four 220-330 resistors for the LED's, and cables.

Part 2: Online Simulation / Diagram

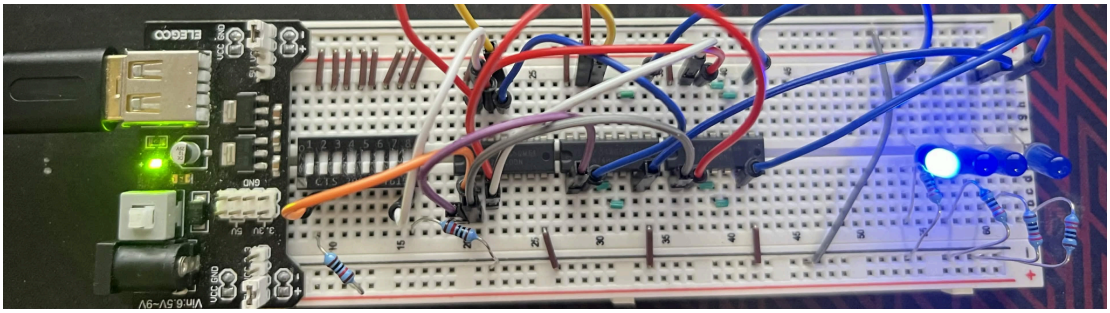
https://www.falstad.com/s.php?s=F7ADkS.Using Falstad I prepared a simulation of what the breadboard will do and the expected outputs for the LED's based on inputs.

To turn on an input you have to press the "L" (0) to turn it into "H" (1).

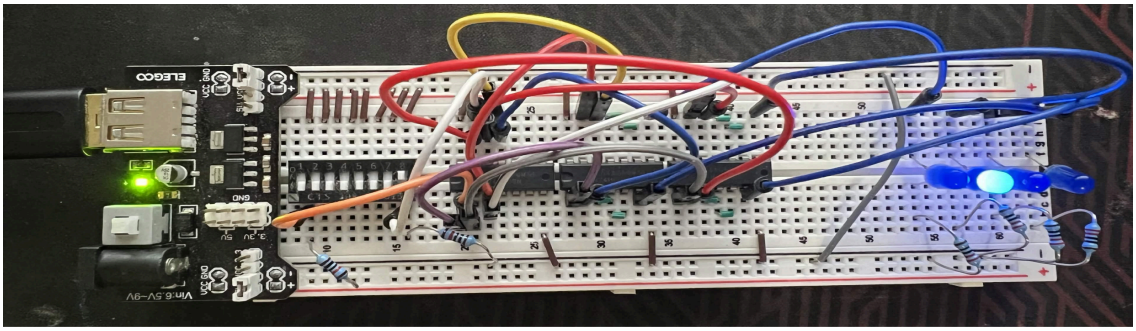


All Cases, Switch 1 = A, Switch 8 = B. LED 1 to 4 : I_0 to I_3

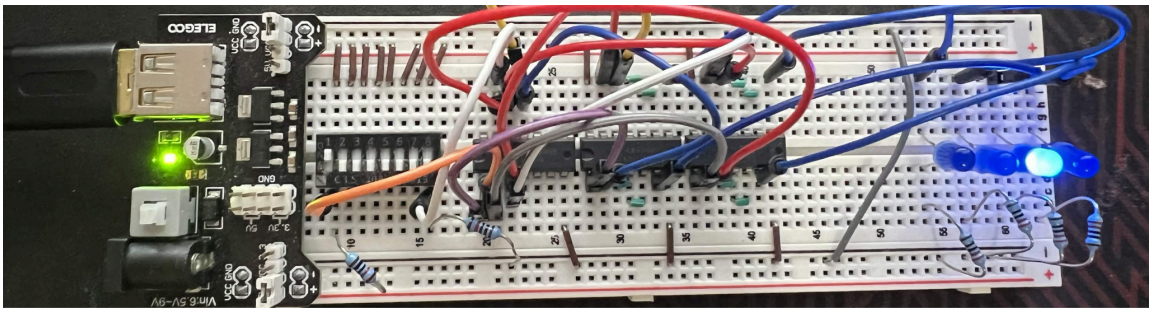
Case 1 (0/0): A=B=0, Output: I_0



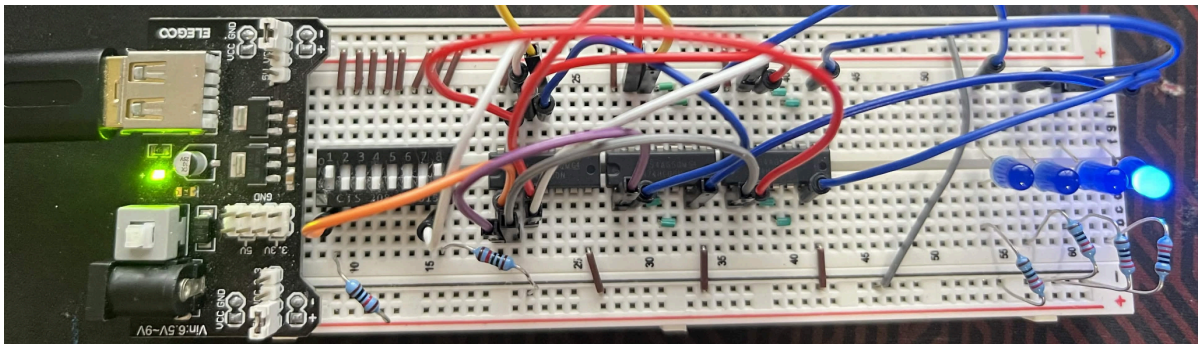
Case 2 (0/1): A=0, B=1 Output: I_1



Case 3 (1/0): A=1, B=0 Output: I_2



Case 4 (1/1): A=B=1 Output: I_3



Conclusion

In conclusion, we successfully designed and simulated a 2-to-4 decoder using only basic NAND gates. The observed outputs matched the expected results for all tested input combinations, confirming that the circuit operates correctly.

This design can also be expanded into larger decoders by adding additional NAND-gate ICs and increasing the number of inputs and outputs.